

VERITAS: The First Dynamic Benchmark for Multimodal Automated Fact-Checking

VERITAS: 首个多模态自动事实核查的动态基准

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Figure 1: images/a0e2caf093ddaaffb90bd8f9788420198d75af88b9a43932689ac115f2f08cf69.jpg

Mar 10, 2026 The video shows the house of Israeli Prime Minister Benjamin Netanyahu on fire.
 2026 年 3 月 10 日视频显示以色列总理本雅明·内塔尼亚胡的房屋正在燃烧。



Немецкий сатирический журнал Titanic выпустил номер с обложкой, на которой Владимир Зеленский изображён в виде чёрной дыры, поглощающей деньги и вооружение.
 Mar 04, 2025 德国讽刺杂志《Titanic》发行了一期特刊，封面将弗拉基米尔·泽连斯基描绘成一个吞噬金钱和军备的黑洞。

德国讽刺杂志《Titanic》发行了一期特刊，封面将弗拉基米尔·泽连斯基描绘成一个吞噬金钱和军备的黑洞。Mar 04, 2025 特朗普指控莫迪在印度使用电子计票机作弊以赢得选举。



The video shows a helicopter intentionally starting the 2023 wildfires in Canada.

视频显示一架直升机故意引发了加拿大 2023 年的山火。



Gambar ini adalah citra buatan AI yang menampilkan Vladimir Putin dan Narendra Modi berdoa di Kuil Mahakaleshwar, Ujjain.

这张图片是人工智能生成的图像，展示了弗拉基米尔·普京和纳伦德拉·莫迪在乌贾因的马哈卡莱什瓦尔神庙中祈祷。



El video muestra una protesta anticorrupción en Belgrado.

视频展示了贝尔格莱德的一场反腐败抗议活动。

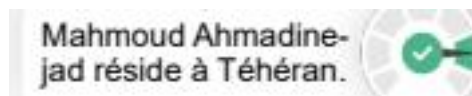


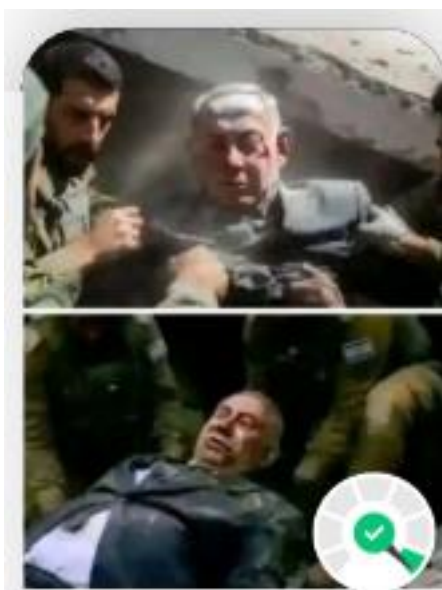
Figure 2: images/91588e557431208ba6dd9f20e078eb5acc4b7b4d2edad9105149743e03c0ceec.jpg



Figure 3: images/76f3921df1eed74a16d95a5544b1f6d9facb50ccbbcfde54f3555be2ed65a884.jpg



Figure 4: images/ef112b6d3229becd88e771a9032317a51d46249ebd65fd39931c70444d369b2c.jpg



These images are generated by artificial intelligence, showing the victims of the Iranian missile attack.

Israel's former prime minister Netanyahu said these images were generated by artificial intelligence, showing the victims of the Iranian missile attack.



图片展示了被分配到月球飞越任务的 NASA 阿耳忒弥斯 II 航天员。视频显示了一架直升机故意在加拿大引发了 2023 年的野火。

印度旁遮普邦的教育标准正在下降。



၀၀ ၀၀၀“”၀၀၀၀၀ ၀၀၀၀၀ ၀၀ ၀၀၀၀ ၀၀၀၀၀၀၀၀၀ ၀၀၀၀၀၀ ၀၀၀၀၀၀ ၀၀၀၀၀၀၀၀ ho၀ ၀၀ ta၀၀၀၀၀၀၀၀၀
၀၀၀၀၀၀၀ ၀၀၀ ၀၀၀၀၀၀၀၀ ၀၀၀၀ ၀၀ ၀၀၀၀ ၀၀၀၀၀ ၀၀

警方清场占领的北莱茵-威斯特法伦州卢特泽拉特村庄的行动耗费了大约 2500 万欧元。

Mar 4, 2025

发芽刺激导致脑出血
预防·治疗方法不是

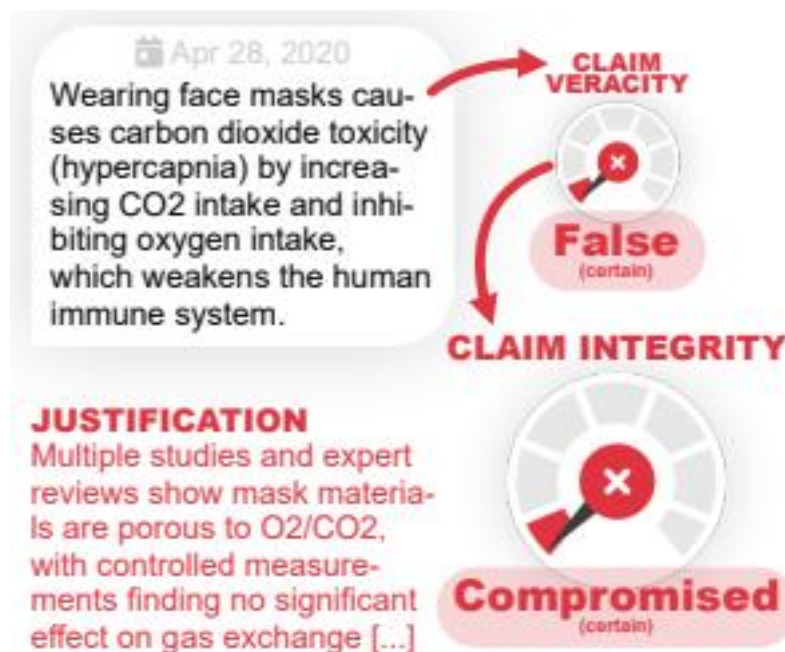


Figure 5: images/3a725ebec8188b9dbb9814e7b48cbdae76794f5a00b8e04fe8bd9bd3f0608570.jpg



Figure 6: images/48b83b065fb218cf0d75452fa13837dc602d5af9e8e236c1cb83daf74414bde4.jpg

JUSTIFICATION The clip was posted to TikTok on Oct 21, 2022, with a [matching description ...]; however, the video's exact origin could not be independently verified, so some uncertainty remains.

说明该视频于 2022 年 10 月 21 日发布在 TikTok 上，配有 [匹配的描述 ...]；然而，该视频的确切来源无法独立核实，因此仍存在一些不确定性。

Figure 1: The VERITAS benchmark: Example claims including media, claim date, and claim integrity score. The two claims on the right showcase lower-level annotations of media/claim properties used to infer the overall integrity. Each annotation contains a justification. New claims are

added quarterly via an automated pipeline.

图 1: VERITAS 基准测试: 包含媒体、声明日期和声明完整性评分的示例声明。右边的两个声明展示了用于推断整体完整性的媒体/声明属性的低级别注释。每个注释都包含一个理由。新的声明通过自动化流程每季度添加。

Abstract

摘要

The growing scale of online misinformation urgently demands Automated Fact-Checking (AFC). Existing benchmarks for evaluating AFC systems, however, are largely limited in terms of task scope, modalities, domain, language diversity, realism, or coverage of misinformation types. Critically, they are static, thus subject to data leakage as their claims enter the pretraining corpora of LLMs. As a result, benchmark performance no longer reliably reflects the actual ability to verify claims. We introduce Verified Theses and Statements (VERITAS), the first dynamic benchmark for multimodal AFC, designed to remain robust under ongoing large-scale pretraining of foundation models. VERITAS currently comprises 25,000 real-world claims from 104 professional fact-checking organizations across 54 languages, covering textual and audiovisual content. Claims are added quarterly via a fully automated seven-stage pipeline that nor-

在线虚假信息规模的扩大迫切需要自动事实核查 (AFC)。然而, 现有的用于评估 AFC 系统的基准主要在任务范围、模态、领域、语言多样性、现实性或虚假信息类型覆盖方面存在局限。关键的是, 这些基准是静态的, 因此当它们的声明进入大型语言模型 (LLMs) 的预训练语料库时, 容易导致数据泄露。因此, 基准性能不再可靠地反映验证声明的实际能力。我们引入了 Verified Theses and Statements (VERITAS), 这是首个针对多模态 AFC 的动态基准, 旨在在基础模型持续进行大规模预训练的情况下保持稳健性。VERITAS 目前包含来自 54 种语言、104 个专业事实核查组织的 25,000 个真实世界声明, 涵盖文本和视听内容。声明通过一个完全自动化的七阶段流程每季度添加, 该流程不涉及人工干预。

malizes claim formulation, retrieves original media, and maps heterogeneous expert verdicts to a novel, standardized, and disentangled scoring scheme with textual justifications. Through human evaluation, we demonstrate that the automated annotations closely match human judgments. We commit to updating VERITAS in the future, establishing a leakage-resistant benchmark, supporting meaningful AFC evaluation in the era of rapidly evolving foundation models. The code and data are publicly available under veritas.mai.informatik.tu-darmstadt.de.

最小化主张的形成, 检索原始媒体, 并将异构专家裁决映射到一种新颖、标准化且解耦的评分方案, 并附有文本证明。通过人工评估, 我们证明了自动标注与人类判断高度一致。我们承诺在未来更新 VERITAS, 建立一个抗泄漏的基准, 支持在基础模型快速演进时代有意义的 AFC 评估。代码和数据可在 veritas.mai.informatik.tu-darmstadt.de 上公开获取。

1 Introduction

1 引言

Mis- and disinformation lead the current short-term risks for global society (Elsner et al., 2025). Humans perceive multimodal information as more credible than text alone (Newman et al., 2012), making images and videos powerful tools for persuasion (Hameleers et al., 2020; Greifeneder et al., 2020). As a result, multimodal content achieves

错误信息和虚假信息导致了当前全球社会的短期风险 (Elsner 等, 2025)。人类认为多模态信息比单独的文字更具可信度 (Newman 等, 2012), 这使得图像和视频成为强有力的说服工具 (Hameleers 等, 2020; Greifeneder 等, 2020)。因此, 多模态内容实现了

higher engagement and spreads faster, particularly in the context of misinformation (Li and Xie,

2020; Zannettou et al., 2018; Wang et al., 2021). Du-four et al. (2024) found that visual misinformation accounts for up to 80% of fact-checked claims. At the same time, generative methods like NANO BANANA (Google, 2025b) or SORA 2 (OpenAI, 2025b) evolve rapidly, producing realistic deep-fakes, outpacing countermeasures. Manual fact-checking, however, is a very laborious task (Hassan et al., 2015), which gives rise to robust multimodal Automated Fact-Checking (AFC).

更高的参与度和传播速度，尤其是在错误信息的背景下 (Li 和 Xie, 2020; Zannettou 等, 2018; Wang 等, 2021)。Du-four 等 (2024) 发现，视觉类错误信息占事实核查声明的最高比例可达 80%。同时，像 NANO BANANA (Google, 2025b) 或 SORA 2 (OpenAI, 2025b) 这样的生成方法迅速发展，能够产生逼真的深度伪造内容，超越了对抗措施。然而，手动事实核查是一项非常繁琐的任务 (Hassan 等, 2015)，这促使了稳健的多模态自动事实核查 (AFC) 的发展。

To evaluate AFC methods, numerous benchmarks have been proposed. However, many lack full modality coverage, missing images or videos (Schlichtkrull et al., 2023; Cao et al., 2025; Geng et al., 2025), restrict the task scope to isolated subproblems (Papadopoulos et al., 2024; Tonglet et al., 2024), or rely heavily on synthetically generated claims (Xu et al., 2024a, 2025), raising concerns about realism. Additionally, practically all AFC benchmarks model the task as a single-label classification problem, conflating orthogonal claim properties into coarse labels, often ignoring uncertainty and finer-grained properties, and treating severe confusions (True vs. False) equally to less severe ones (e.g., True vs. Not Enough Information).

为了评估 AFC 方法，已经提出了许多基准测试。然而，许多基准测试缺乏完整的模态覆盖，缺少图像或视频 (Schlichtkrull 等, 2023; Cao 等, 2025; Geng 等, 2025)，将任务范围限制在孤立的子问题 (Papadopoulos 等, 2024; Tonglet 等, 2024)，或严重依赖合成生成的声明 (Xu 等, 2024a, 2025)，这引发了对现实性的担忧。此外，几乎所有 AFC 基准测试都将任务建模为单标签分类问题，将相互独立的声明属性合并到粗略的标签中，通常忽略了不确定性以及更细粒度的属性，并且将严重的混淆 (真实 vs. 错误) 与较不严重的混淆 (如真实 vs. 信息不足) 同等对待。

More fundamentally, all existing AFC benchmarks are static and therefore vulnerable to data leakage. State-of-the-art AFC systems rely on large language models (LLMs) that are continually pretrained on public web data. Since benchmark claims originate from the same public sources, models may implicitly encode both claims and verdicts in their parametric knowledge, undermining meaningful evaluation. We empirically substantiate this effect in Fig. 5, demonstrating a performance drop for claims published after the models’ knowledge cutoff date.

更根本地说，所有现有的 AFC 基准测试都是静态的，因此容易受到数据泄露的影响。最先进的 AFC 系统依赖于在公共网络数据上持续预训练的大型语言模型 (LLMs)。由于基准测试的声明来源于相同的公共来源，模型可能在参数知识中隐式地编码了声明和判决，从而破坏了有意义的评估。我们在图 5 中实证地证明了这种影响，展示了在模型知识截止日期之后发布的声明所带来的性能下降。

To address these limitations, we introduce Verified Theses and Statements (VERITAS), the first dynamic AFC benchmark. VERITAS is extended quarterly with newly emerging real-world claims, effectively mitigating data leakage in future splits through fully automated data acquisition. It is also the first benchmark to combine real-world multimodal claims featuring both images and videos. Unlike the vast majority of previous work, VERITAS also provides multilingual, open-domain, and balanced claim data with verdict annotations accompanied by textual justifications for the rulings, derived from expert ground truth. Fig. 1 shows example claims from VERITAS.

为了解决这些限制，我们引入了验证性论文和声明 (Verified Theses and Statements, 简称 VERITAS)，这是首个动态的 AFC 基准测试集。VERITAS 会定期扩展，加入新出现的现实世界声明，通过全自动的数据采集有效缓解未来分割中的数据泄露问题。它也是首个结合现实世界多模态声明的基准测试集，这些声明同时包含图像和视频。与大多数先前的工作不同，VERITAS 还提供了多语言、开放领域且平衡的声明数据，并附有判决注释，这些注释包含文本形式的判决理由，来源于专家的真实数据。图 1 展示了 VERITAS 中的示例声明。

Moreover, we contribute a novel rating scheme that (1) disentangles previously coarse-grained rulings into separate fine-grained claim and media properties, (2) incorporates uncertainty by modeling each of these properties on a scale from -1 to 1 allowing to (3) employ Mean Squared Error (MSE) as an evaluation metric to reward near-correct predictions and strongly penalize

flipped decisions.

此外，我们贡献了一种新颖的评分方案，该方案具有以下特点：(1) 将先前粗粒度的裁定分解为独立的细粒度主张和媒体属性；(2) 通过在 -1 到 1 的范围内对这些属性进行建模，来纳入不确定性，从而允许 (3) 使用均方误差 (MSE) 作为评估指标，以奖励接近正确的预测，并对翻转的决策施加严厉的惩罚。

Finally, we benchmark state-of-the-art multimodal AFC systems and strong LLM baselines, revealing substantial room for improvement. We made the benchmark data and pipeline code available at veritas.mai.informatik.tu-darmstadt.de and commit to extending VERITAS in each subsequent quarter until at least Q4 2028.

最后，我们对最先进的多模态 AFC 系统和强大的 LLM 基线进行了基准测试，揭示了仍有显著的改进空间。我们将在 veritas.mai.informatik.tu-darmstadt.de 上公开基准数据和流水线代码，并承诺在至少 2028 年第四季度之前每个后续季度都扩展 VERITAS。

2 Related Work

2 相关工作

Benchmarks by Task Scope and Domain Automated Fact-Checking (AFC) remains a challenging and unsolved problem, particularly in realistic, open-domain settings (Akhtar et al., 2023; Dmonte et al., 2025; Schlichtkrull et al., 2024). To tackle the complexity, most benchmarks restrict the task scope, either by decomposing AFC into sub-tasks, such as image contextualization (Tonglet et al., 2024), out-of-context detection (Xu et al., 2024b; Papadopoulos et al., 2024; Luo et al., 2021; Tonglet et al., 2025; Aneja et al., 2021), manipulation detection (Shao et al., 2023), check-worthiness estimation (van der Meer et al., 2025), fact-check retrieval (Papadopoulos et al., 2025), content interpretation (Jin et al., 2024), deepfake detection (Skoularikis et al., 2025), claim disambiguation (Staliunaite and Vlachos, 2025; Glockner et al., 2024), claim detection (Cheema et al., 2022), claim normalization (Sundriyal et al., 2023), or claim matching (Pisarevskaya and Zubiaga, 2025). Some benchmarks limit the domain to specific platforms like Reddit (Nakamura et al., 2020), or topics like elections (Khatiwada et al., 2025), finance (Rangapur et al., 2025), climate charts (Su et al., 2025), or the Ukraine-Russia war (Bondielli et al., 2024). In contrast, VERITAS targets the holistic and open-domain, end-to-end task of claim verification, where the goal is to predict expert ratings.

按任务范围和领域划分的基准测试，自动事实核查 (AFC) 仍然是一个具有挑战性且尚未解决的问题，特别是在现实、开放领域的情境下 (Akhtar 等, 2023; Dmonte 等, 2025; Schlichtkrull 等, 2024)。为了解决这种复杂性，大多数基准测试通过限制任务范围来简化问题，例如将 AFC 分解为子任务，如图像语境化 (Tonglet 等, 2024)，上下文不一致检测 (Xu 等, 2024b; Papadopoulos 等, 2024; Luo 等, 2021; Tonglet 等, 2025; Aneja 等, 2021)，操纵检测 (Shao 等, 2023)，可核查性估计 (van der Meer 等, 2025)，事实核查检索 (Papadopoulos 等, 2025)，内容解释 (Jin 等, 2024)，深度伪造检测 (Skoularikis 等, 2025)，声明歧义消解 (Staliunaite 和 Vlachos, 2025; Glockner 等, 2024)，声明检测 (Cheema 等, 2022)，声明规范化 (Sundriyal 等, 2023)，或声明匹配 (Pisarevskaya 和 Zubiaga, 2025)。一些基准测试将领域限制在特定平台，如 Reddit (Nakamura 等, 2020)，或特定主题，如选举 (Khatiwada 等, 2025)，金融 (Rangapur 等, 2025)，气候图表 (Su 等, 2025)，或乌克兰-俄罗斯战争 (Bondielli 等, 2024)。相比之下，VERITAS 目标是全面且开放领域的端到端声明验证任务，其目标是预测专家评分。

Benchmarks by Modality and Annotation. Early AFC benchmarks are predominantly text-only, including FEVER (Thorne et al., 2018) and LIAR (Wang, 2017), with later publications such as AVERITEC (Schlichtkrull et al., 2023) address-

按模态和注释的基准测试。早期的 AFC 基准测试主要以文本为主，包括 FEVER (Thorne 等, 2018) 和 LIAR (Wang, 2017)，而后来的出版物如 AVERITEC (Schlichtkrull 等, 2023) 则涉及——

Benchmark	Primary task												
Images	Videos	Dynamics	Automated	Real claims	Misinfo coverage # Languages	Open-domain	Justification	Balance	# Instances				
Annotation													
M^3A (Xu et al., 2024a)	CV	□	□	-	□	-	-	1	News	-	-	-	7.3 M
MDA^3 -DB (Mu et al., 2025)	CV	□	□	-	□	-	-	1	News	-	-	-	90 K
XFACTA (Xiao et al., 2025)	CV	□	-	□	-	□	□	1	SM	-	□	-	2.4 K
MUMIN (Nielsen and McConville, 2022)	CV	□	-	-	□	□	□	41	□	-	-	-	13 K
KHATIWADA ET AL. (Khatiwada et al., 2025)	CV	□	-	-	□	□	□	1	SM	-	-	-	77 K
CLAIMREVIEW (Ren et al., 2025)	CV	2024+ □	-	-	-	□	□	1	□	-	-	-	300
M4FC (Geng et al., 2025)	CV	□	-	-	-	□	□	10	□	-	□	-	7.0 K
REALFACTBENCH (Vang et al., 2025b)	CV	□	-	-	-	□	□	1	□	-	-	-	6.0 K
MR^2 (Hu et al., 2023)	CV	□	-	-	-	□	□	2	□	-	-	-	15 K

VLD-BENCH	Ka CV	□	-	-	-	□	□	1	News	-	□	63 K
(Kha et al., 2025)												
AVERIMATEC	CV	□	-	-	-	□	-	1	□	□	-	1.3 K
(Kha et al., 2025)												
FACTIFY	Mishra CV	□	-	-	-	□	□	1	□	-	□	50 K
(Mishra et al., 2022)												
FACTIFY 3M	CV	□	-	-	-	□	□	1	□	-	-	3 M
(Chakraborty et al., 2023)												
MMFAKEBENCH	CV	□	-	-	-	-	□	1	□	-	-	11 K
(Kha et al., 2025)												
OMNIFAKE	CV	□	-	-	-	-	-	1	SM	-	-	127 K
(Kha et al., 2025a)												
TRUE (Niu et al., 2025)	CV	-	□	-	□	□	□	1	□	□	-	2.9 K
GROUNDING-360	CV	-	□	-	-	□	□	1	□	-	□	2.0 K
(Niu et al., 2025a)												
MMOOC (Xu et al., 2024b)	OOCD	□	□	-	□	-	-	1	News	-	-	455 K
VERITE (Papadopoulos et al., 2024)	OOCD	□	-	-	□	□	-	1	□	-	□	1.0 K
NEWSCLIPPINGOOD	OOCD	□	-	-	□	□	-	1	News	-	□	988 K
(Luo et al., 2021)												

5PILS- OOC (Ton- glet et al., 2025)	OOCD	□	-	-	-	□	-	1	□	-	□	1.2 K
COSMOS (test split) (Aneja et al., 2021)	OOCD	□	-	-	-	□	-	1	News	-	□	1.7 K
<i>DGM</i> ⁴ (Shao et al., 2023)	MD	□	-	-	□	-	□	1	News	-	-	230 K
VERITAS (Ours)	CV	□	□	□	□	□	□	54+	□	□	□	25 K+

基准 测试 图片	主要 任务 视频	动态	自动 化	真实 主张	Misinfo cov- er- age # 语言	开放 领域	理由	平衡	# 实 例			
注释												
<i>M³A</i> (Xu 等, 2024a)	简历	□	□	-	□	-	-	1	新闻	-	-	7.3 M
<i>MDA³</i> -DB (Xu 等, 2025)	简历	□	□	-	□	-	-	1	新闻	-	-	90 K
XFACTA (Xiao 等, 2025)	简历	□	-	□	-	□	□	1	SM	-	□	2.4 K
MUMIN (Nielsen 和 Mc- Conville, 2022)	简历	□	-	-	□	□	□	41	□	-	-	13 K

Khatiwada 等人 (Khati- wada 等 人, 2025 年)	简历	□	-	-	□	□	□	1	SM	-	-	77 K
CLAIM REVIEW 等, 2025)	REVIEW 2024+	□	-	-	-	□	□	1	□	-	-	300

M4FC	(Geng 等, 2025)	CV (Cur- ricu- lum Vi- tae) 是用 于求 职时 展示 个人 经 历、 技能 和教 育背 景的 文 件。 它通 常包 括以 下内 容：- 个人 信息： 姓 名、 联系 方 式、 地址 等。- 教育 背景： 所获 得的 学 位、 学校 名 称、 学习 时间 等。- 工作 经历： 担任 过的 职 位、 公司 名 称、 工作 时间 等。- 技能： 语言 能	□	-	-	-	□	□	10	□	-	□	7.0 K
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REALFAVORABLE CV (Vaidya et al., 2025b)		□	-	-	-	□	□	1	□	-	-	6.0 K
MR^2 (Hu et al., 2023)	简历	□	-	-	-	□	□	2	□	-	-	15 K
VLD-BENCH (Khalil et al., 2025)	简历	□	-	-	-	□	□	1	新闻	-	□	63 K
AVERIM (Alec et al., 2025)	CV	□	-	-	-	□	-	1	□	□	-	1.3 K
事实化 (Mishra et al., 2022)	CV	□	-	-	-	□	□	1	□	-	□	50 K
FACTIFY 3M (Chakraborty et al., 2023)	简历	□	-	-	-	□	□	1	□	-	-	3 M
MMFAKE-BENCH (Khalil et al., 2025)	简历	□	-	-	-	-	□	1	□	-	-	11 K
OMNIFAKE (Ali et al., 2025a)	CV	□	-	-	-	-	-	1	SM	-	-	127 K
TRUE (Niu et al., 2025)	简历	-	□	-	□	□	□	1	□	□	-	2.9 K
GROUNDING-36 (Yang et al., 2025a)	简历	-	□	-	-	□	□	1	□	-	□	2.0 K
MMOOC (Xu et al., 2024b)	OOCD	□	□	-	□	-	-	1	新闻	-	-	455 K
VERITE (Papadopoulos et al., 2024)	OOCD	□	-	-	□	□	-	1	□	-	□	1.0 K
新闻剪辑 (Luo et al., 2021)	OOCD	□	-	-	□	□	-	1	新闻	-	□	988 K

5PILS- OOC (Ton- glet et al., 2025)	OOCD	□	-	-	-	□	-	1	□	-	□	1.2 K
COSMOS (test split) (Aneja 等 人, 2021 年)	OOCD	□	-	-	-	□	-	1	新闻	-	□	1.7 K
<i>DGM</i> ⁴ (邵 等, 2023)	MD	□	-	-	□	-	□	1	新闻	-	-	230 K
VERITAS (Ours)	简历	□	□	□	□	□	□	54+	□	□	□	25 千以上

Table 1: Overview of related multimodal AFC benchmarks. Primary task: CV = Claim Verification, OOOD = Out-of-Context Detection, MD = Manipulation Detection. Dynamic: Whether the authors highlight the benchmark as extensible with new claims (□) and if the authors committed to update it regularly (□). Automated: Whether the construction pipeline is fully autonomous. Real claims: Whether the benchmark is entirely made from real-world claims (and perhaps close derivations) (□) or contains some (□) or mostly (-) synthetic, i.e., invented claims. Misinfo coverage: Whether the claims cover the full (□) or broad (□) spectrum of contemporary misinformation types for the given modalities, or just selected types (-). Open-domain: SM = Social Media. Justifications: Whether the benchmark contains explanations in addition to the annotations.

表 1: 相关多模态 AFC 基准概述。主要任务: CV = 声明验证, OOOD = 上下文外检测, MD = 操纵检测。动态: 作者是否将基准标记为可扩展 (□) 以及是否承诺定期更新 (□)。自动化: 构建流程是否完全自动化。真实声明: 基准是否完全由现实世界声明 (以及可能的近似推导) 组成 (□), 或者包含一些 (□) 或主要 (-) 合成声明, 即编造的声明。错误信息覆盖范围: 声明是否覆盖了给定模态下当代错误信息类型的全部 (□) 或广泛 (□) 范围, 或者只是部分 (-)。开放领域: SM = 社交媒体。解释: 基准是否包含除标注之外的解释。

ing previous issues like temporal leakage (Glockner et al., 2022). However, only multimodal benchmarks can capture the important role of visual content—see Tab. 1 for a detailed comparison. We consider multimodal AFC benchmarks to incorporate claims with associated images and/or videos, unlike datasets that use multimodal evidence but retain text-only claims as proposed by Tang et al. (2024); Yao et al. (2023); Wang et al. (2025). Multimodal AFC benchmarks are often limited in size (Zlatkova et al., 2019), or miss either images or videos entirely. Only a few datasets support both images and videos (Xu et al., 2024a, 2025). In contrast, VERITAS covers all common modalities. Additionally, among all multimodal AFC benchmarks, only a few provide justifications for explanation (Cao et al., 2025; Niu et al., 2025). Almost all benchmarks use a single-label n-class annotation scheme, often entangling different, orthogonal properties (such as media authenticity and claim veracity) in the same coarse category with unclear separation. The only exception is Khatiwada et al. (2025) who employ a multi-labeling scheme. VERITAS explicitly models claim and media properties

解决诸如时间泄漏 (Glockner 等, 2022) 等先前问题。然而, 只有多模态基准测试能够捕捉视觉内容的重要作用—参见表 1 进行详细比较。我们认为多模态 AFC 基准测试应包含与图像和/或视频相关的声明,

不同于 Tang 等人 (2024); Yao 等人 (2023); Wang 等人 (2025) 所提出的使用多模态证据但保留纯文本声明的数据集。多模态 AFC 基准测试通常在规模上受到限制 (Zlatkova 等, 2019), 或者完全缺少图像或视频。只有少数数据集支持图像和视频 (Xu 等, 2024a, 2025)。相比之下, VERITAS 覆盖了所有常见的模态。此外, 在所有多模态 AFC 基准测试中, 只有少数提供解释的依据 (Cao 等, 2025; Niu 等, 2025)。几乎所有基准测试都使用单一标签的 n 类注释方案, 经常将不同的、正交的属性 (如媒体真实性与声明真实性) 混杂在同一粗略类别中, 而分离不清晰。唯一的例外是 Khatiwada 等人 (2025), 他们采用了多标签方案。VERITAS 明确建模声明和媒体属性

with disentangled, uncertainty-aware scores and textual justifications.

具有解耦的、不确定性感知的分数和文本解释。

Dynamic Claim Datasets. Dynamic benchmarking has mainly been explored in adversarial settings where data collection and model training co-evolve (Shirali et al., 2023), most prominently in DYNABENCH (Kiela et al., 2021). By contrast, VERITAS is updated independently of evaluated models. Sustaining a dynamic dataset requires automation: Many benchmarks rely on human annotation (Xiao et al., 2025) or synthetic claim generation (Xu et al., 2024a, 2025), raising concerns about scalability or realism. MUMIN (Nielsen and McConville, 2022) and TRUE (Niu et al., 2025) are the only ones to accomplish both realism and automation. However, MUMIN lacks videos, and TRUE lacks images. Two dynamic claim repositories exist: CLAIMSKG (Tchechmedjiev et al., 2019) and CIMPLEKG (Burel et al., 2025). Both aggregate fact-checked claims daily, but are text-only and not designed as evaluation benchmarks. Xiao et al. (2025) express intent to “continuously update” their benchmark, XFACTA, yet the most recent change is more than

动态索赔数据集。动态基准测试主要在对抗性设置中进行探索, 其中数据收集和模型训练相互演化 (Shirali 等, 2023), 最显著的例子是 DYNABENCH (Kiela 等, 2021)。相比之下, VERITAS 是独立于被评估模型进行更新的。维持一个动态数据集需要自动化: 许多基准测试依赖于人工标注 (Xiao 等, 2025) 或合成索赔生成 (Xu 等, 2024a, 2025), 这引发了关于可扩展性或现实性的担忧。MUMIN (Nielsen 和 McConville, 2022) 和 TRUE (Niu 等, 2025) 是唯一实现现实性和自动化的系统。然而, MUMIN 缺乏视频, TRUE 缺乏图像。存在两个动态索赔存储库: CLAIMSKG (Tchechmedjiev 等, 2019) 和 CIMPLEKG (Burel 等, 2025)。两者每天汇总经过事实核查的声明, 但都是纯文本形式, 并未设计为评估基准。Xiao 等 (2025) 表达了“持续更新”其基准 XFACTA 的意愿, 但最近一次更改已超过

8 months ago. Fatahi Bayat et al. (2025) propose a dynamic, text-only benchmark that consists not of claims but of prompts of varying complexity to probe the factuality of LLMs.

8 个月前。Fatahi Bayat 等人 (2025) 提出了一种动态的纯文本基准测试, 该基准测试不包含声明, 而是包含复杂程度各异的提示, 以探测大型语言模型的事实性。

In a nutshell, and as can be seen in Tab. 1, prior work always lacks multiple important properties. In contrast, VERITAS combines real-world multimodal claims, fine-grained annotations derived from expert-provided rulings with justifications, and a fully automated construction pipeline. Its dynamic nature makes VERITAS the first benchmark specifically designed for long-term, leakage-resistant evaluation of multimodal AFC systems.

简而言之, 如表 1 所示, 以往的工作总是缺乏多个重要的特性。相比之下, VERITAS 结合了真实世界中的多模态声明、来自专家裁定的细粒度标注及其理由, 并且拥有完全自动化的构建流程。其动态特性使得 VERITAS 成为首个专门设计用于长期、防泄漏评估的多模态 AFC 系统的基准。

3 The VERITAS Construction Pipeline

3 VERITAS 构建流水线

VERITAS (Verified Theses and Statements) is a dynamic, multimodal, and multilingual benchmark for evaluating Multimodal Automated Fact-Checking (MAFC) systems. It gets extended with recent, real-world claims each quarter, combining text, images, and videos. The data is gathered via a seven-stage pipeline enabling large-scale, standardized claim acquisition and annotation.

VERITAS（验证的论点和声明）是一个动态的、多模态的、多语言的基准，用于评估多模态自动事实核查（MAFC）系统。它每季度都会加入最新的、现实世界的声明，结合文本、图片和视频。数据是通过一个七阶段的流程收集的，使得大规模、标准化的声明获取和标注成为可能。

LLMs have proven to be reliable enough for automated annotation (Khatriwada et al., 2025). Across stages, we employ LLMs from the GPT 5 (OpenAI, 2025a) and GEMINI (Google, 2025a) families as these represent the state of the art in multimodal language modeling at the time of data collection, achieving leading performance in image and video understanding tasks, respectively. We select model sizes based on task complexity and model availability (see App. A for the used LLM versions). For multi-step reasoning, we apply chain-of-thought prompting; for generation tasks, we use few-shot in-context examples to constrain output style. Please refer to the code release for the prompts. Fig. 2 summarizes the seven autonomous stages we are going to describe next—App. B contains additional details.

大型语言模型（LLMs）已被证明在自动化标注方面足够可靠（Khatriwada 等，2025）。在各个阶段中，我们使用来自 GPT 5（OpenAI, 2025a）和 GEMINI（Google, 2025a）系列的 LLMs，因为这些模型在数据收集时代表了多模态语言建模的最先进技术，在图像和视频理解任务中分别取得了领先的性能。我们根据任务复杂性和模型可用性选择模型的规模（有关使用的 LLM 版本，请参见附录 A）。对于多步推理任务，我们应用链式思维提示；对于生成任务，我们使用少量示例来约束输出风格。请参考代码发布以获取提示。图 2 总结了我们将接下来描述的七个自主阶段——附录 B 包含更多细节。

3.1 Stage 1: Review Discovery

3.1 阶段 1：审查发现

Goal. Retrieve expert-annotated ground truth claims. We automatically collect fact-check reviews via the ClaimReview¹ schema and, where missing, we determine the language of each claim. Output. Parsed ClaimReview data of about 398 K reviews published between Jan 1, 2016 and Mar 31, 2026, including claim text, rating, review URL, date, language, and appearance URLs.

目标。获取专家标注的真实事实声明。我们通过 ClaimReview¹ 架构自动收集事实核查评论，并在缺失时确定每个声明的语言。输出。约 398,000 条事实核查评论的解析数据，这些评论发布于 2016 年 1 月 1 日至 2026 年 3 月 31 日之间，包括声明文本、评分、评论网址、日期、语言和出现网址。

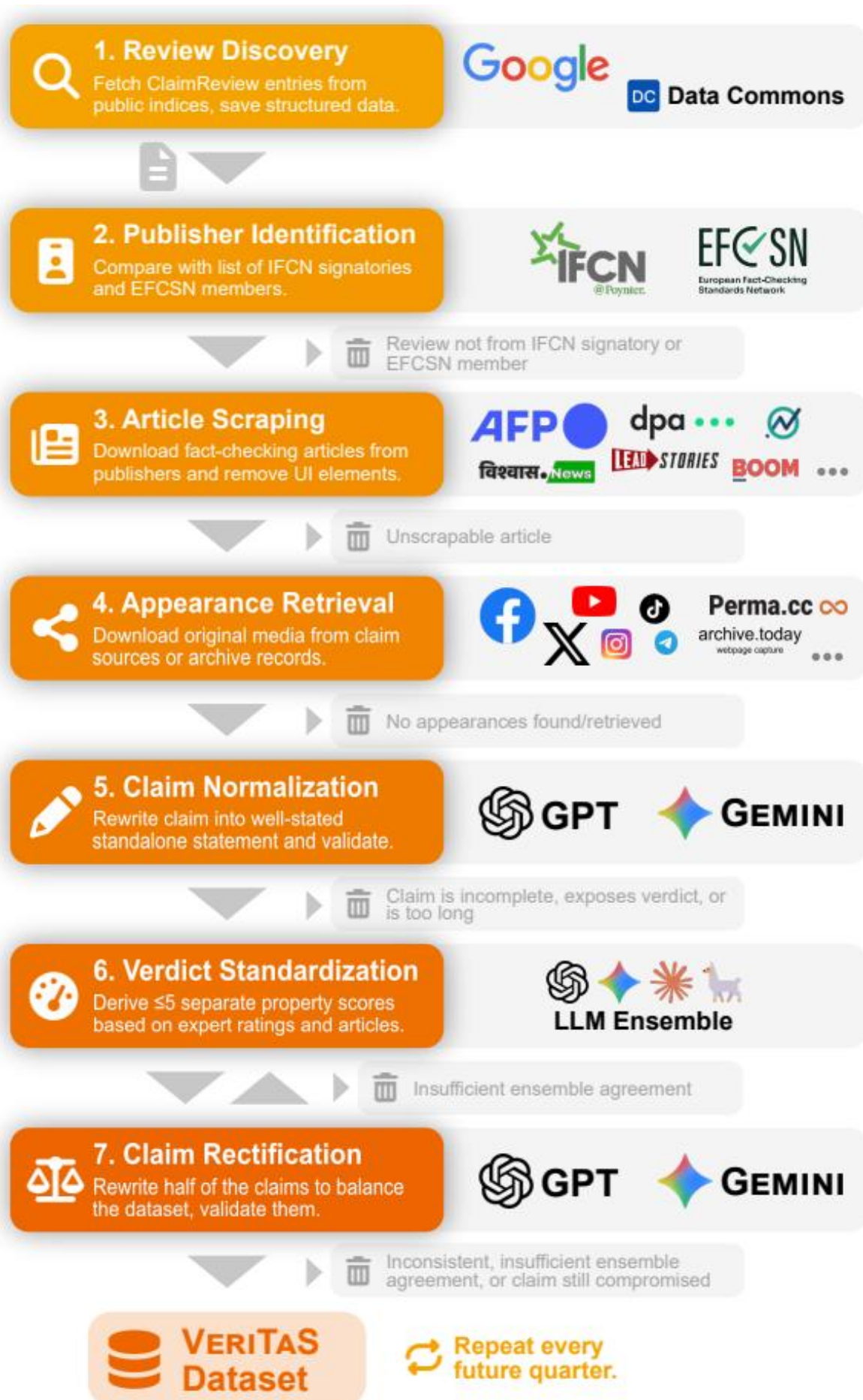


Figure 7: images/d03c8abc8a7509719b849598f434abae395e42109840431eb59b8433468c3877.jpg

Figure 2: The seven stages of VERITAS repeated on a quarterly basis.

图 2：VERITAS 的七个阶段每季度重复一次。

3.2 Stage 2: Publisher Identification

3.2 第二阶段：出版商识别

Goal. Ensure review credibility. We identified 848 distinct publishers via review URLs, and retain reviews only if they originate from credible fact-checkers (cf. App. B), dismissing about 64 K (16%) of the reviews not meeting this criterion. Output. 335 K reviews from professional fact-checking organizations meeting international standards.

目标。确保评论的可信度。我们通过评论链接识别出 848 个不同的出版商，并仅保留来自可信事实核查者的评论（参见附录 B），因此约有 64,000（16%）条不符合此标准的评论被剔除。输出。来自符合国际标准的专业事实核查组织的 335,000 条评论。

3.3 Stage 3: Article Scraping

3.3 阶段 3：文章抓取

Goal. Obtain full fact-check context. Since ClaimReview provides no justifications for the ratings, we scrape the original fact-checking articles incl. media. With GPT-5 NANO we extract the main textual body via span detection, preserving the original text but removing cookie notices, ads, and other UI noise. We discard 8.8 K (4.2%) inaccessible articles as well as 7.3 K (3.5%) unrealistically short articles. Output. 208 K reviews² with cleaned article content.

目标。获取完整的事实核查背景信息。由于 ClaimReview 没有提供评分的依据，我们抓取原始的事实核查文章，包括媒体内容。通过 GPT-5 NANO，我们利用 span 检测提取主要文本内容，在保留原文的基础上，去除 cookie 通知、广告和其他 UI 噪音。我们丢弃了 8.8 K（4.2%）无法访问的文章以及 7.3 K（3.5%）长度不合理的短文章。输出。208 K 条评论²，包含清理后的文章内容。

3.4 Stage 4: Appearance Retrieval

3.4 第四阶段：外观检索

Goal. Recover original claim sources and media. An appearance denotes the original location where a claim surfaced, e.g., a social media post. Claim-Review provides appearance URLs in only 13.3% of cases. For reviews with missing appearances, we prompt GPT-5 MINI to extract appearance URLs from the article text, including archived versions (e.g., Perma.cc; see App. I.2), which are resolved to original sources when possible. If original content cannot be downloaded, archived versions are used. We retain up to two successfully scraped appearances, omitting videos longer than 5 minutes or larger than 128 MB to maintain a reasonable claim scope. 105 K (52.8%) Reviews without any valid appearance are discarded. Output. 94 K reviews² with one or two downloaded appearances.

目标。恢复原始声明来源和媒体。一个出现（appearance）指的是声明首次出现的原始位置，例如社交媒体帖子。Claim-Review 仅在 13.3% 的情况下提供出现的网址。对于缺少出现信息的评论，我们提示 GPT-5 MINI 从文章文本中提取出现的网址，包括存档版本（例如 Perma.cc；参见附录 I.2），当可能时，这些存档版本会解析为原始来源。如果原始内容无法下载，则使用存档版本。我们保留最多两个成功抓取的出现，省略时长超过 5 分钟或大小超过 128 MB 的视频，以保持合理的声明范围。105 K（52.8%）没有有效出现的评论被丢弃。输出。94 K 条评论²带有一个或两个下载的出现。

3.5 Stage 5: Claim Normalization

3.5 阶段 5：索赔规范化

The raw claim text provided by ClaimReview is often malformed, e.g., exposing the verdict, constituting incomplete sentences, or including unnecessary information like “Social media posts claiming...” and omitting media completely. Goal. Produce precise, self-contained claims with relevant media.

ClaimReview 提供的原始声明文本通常格式不正确，例如暴露了判决结果、构成不完整的句子，或者包含不必要的信息如“社交媒体帖子声称.....”，并完全省略了媒体内容。目标：生成精确、自包含的声明，并包含相关媒体。

Media Filtering. We retain only media inherent to the claim, removing duplicates via cosine similarity > 0.85 in the embedding space of the CLIP variant Qdrant/clip-ViT-B-32-vision. Videos are inspected with GEMINI 2.5 FLASH, images with GPT-5 MINI, to exclude irrelevant content. Media precedence follows: original appearance $>$ archived appearance $>$ article media. At most 4 media are kept per claim, prioritizing videos.

媒体过滤。我们仅保留与声明本身相关的媒体内容，在 CLIP 变体 Qdrant/clip-ViT-B-32-vision 的嵌入空间中，通过余弦相似度大于 0.85 来去除重复项。视频内容使用 GEMINI 2.5 FLASH 进行检查，图片内容使用 GPT-5 MINI 进行检查，以排除无关内容。媒体优先级如下：原始出现 $>$ 存档出现 $>$ 文章媒体。每个声明最多保留 4 种媒体，优先选择视频。

Claim Reformulation. Using GEMINI 2.5 PRO and GPT-5.2 for video claims and non-video claims, respectively, raw claim texts are rewritten into about 72 K concise, self-contained statements, conditioned on article content, appearances, and metadata, e.g., claimant and date. Reformulated claims explicitly reference associated media when present and preserve the original language.

声明重述。使用 GEMINI 2.5 PRO 和 GPT-5.2 分别对视频声明和非视频声明进行处理，将原始声明文本重写为约 72,000 条简洁、自包含的陈述，这些陈述基于文章内容、出现情况和元数据（例如声明人和日期）进行条件化生成。重述后的声明在有相关媒体时会明确引用这些媒体，并保留原始语言。

Claim Validation. 480 (0.1%) claims are discarded by LLM validation with GPT-5 MINI and GEMINI 2.5 FLASH as they (i) expose the verdict, (ii) lack required media, or (iii) exceed 70 words to limit claim scope. Output. 72 K validated, well-formed claims² with associated media, if relevant.

声明验证。使用 GPT-5 MINI 和 GEMINI 2.5 FLASH 进行验证时，有 480（0.1%）份声明被丢弃，原因包括：(i) 暴露了判决结果，(ii) 缺少所需的媒体内容，或 (iii) 超过 70 字以限制声明范围。输出。72,000 份经过验证、格式正确的声明²，如有相关媒体内容则一并提供。

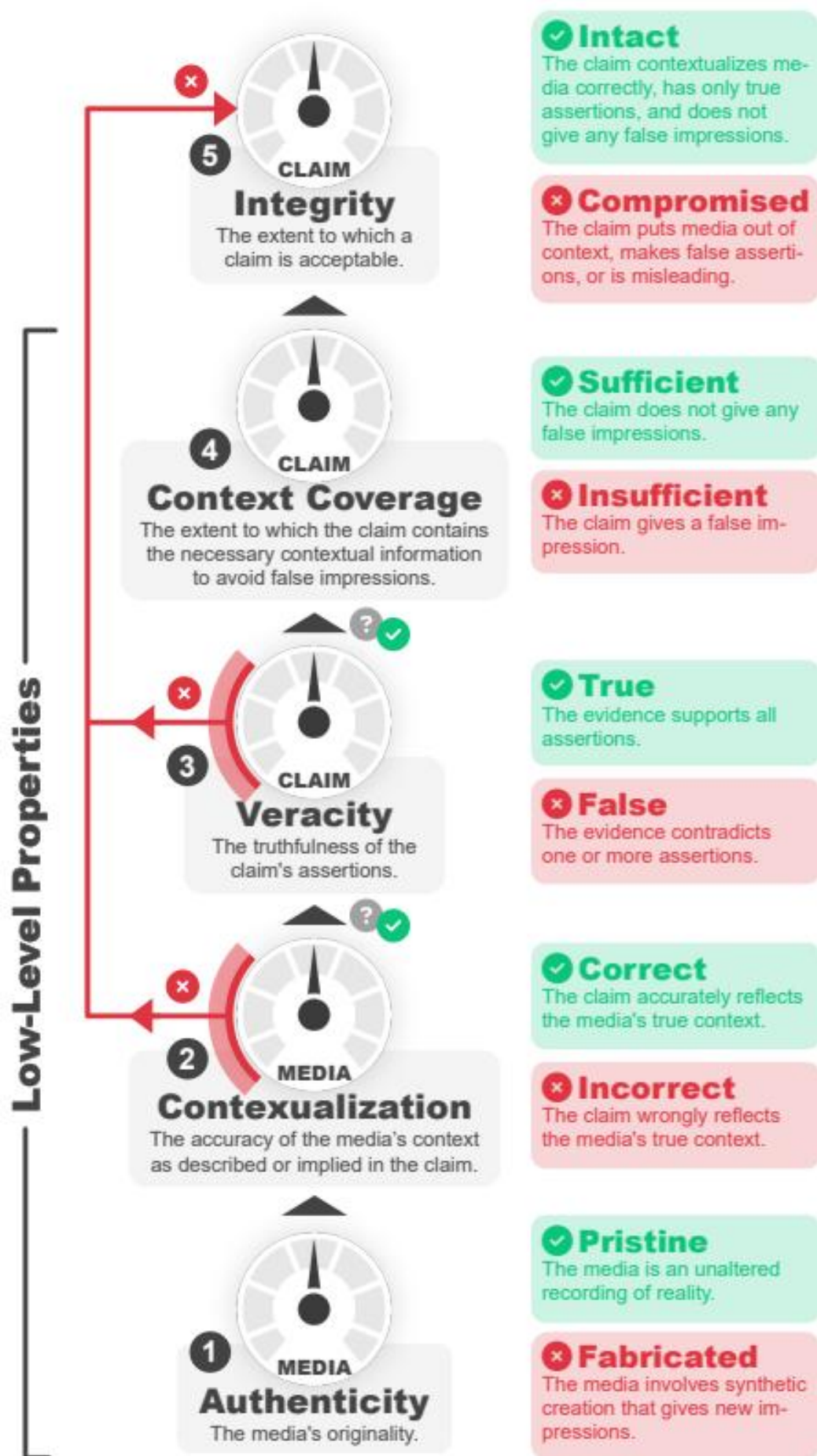


Figure 8: images/445f50616529cecbe432d4c944369bfba98d43b8a59d961d12156ae1e534381c.jpg

Figure 3: Verdict derivation (Stage 6), assessing properties (1) to (4). Negative decisions in (2) or (3) result in early termination. Definitions are shown on the right.

图 3：判决推导（阶段 6），评估属性（1）至（4）。在（2）或（3）中做出负面决定会导致早期终止。右侧展示了定义。

3.6 Stage 6: Verdict Standardization

3.6 阶段 6：裁决标准化

Goal. Standardize heterogeneous fact-check ratings. Fact-checking organizations employ heterogeneous rating schemes that often conflate multiple dimensions, using labels such as “half true,” “legend,” or “three Pinocchios” (see Fig. 10). To obtain a consistent and interpretable representation, we decompose claim assessment into four independent properties that are evaluated sequentially (Fig. 3): (1) Media Authenticity, (2) Media Contextualization, (3) Veracity, and (4) Context Coverage. To retain a single scalar notion of overall acceptability, we further introduce a higher-order property termed Integrity (5). Integrity captures whether a claim is acceptable as presented: Low integrity indicates that at least one lower-level property is violated, whereas high integrity corresponds to positive assessments of Properties (2–4). Integrity constitutes the primary evaluation target, while the lower-level properties serve as explanatory factors that localize the source of compromise and simplify the annotation task. Media-specific properties (1–2) are assessed per media item; if no media is present, evaluation begins at veracity, targeting the textual part. The Context Coverage property explicitly captures

目标。标准化异质性事实核查评分。事实核查组织通常采用异质性评分体系，这些体系常常将多个维度混为一谈，使用诸如“半真半假”、“传说”或“三个匹诺曹”等标签（参见图 10）。为了获得一致且易于理解的表示，我们将声明评估分解为四个独立的属性，依次进行评估（图 3）：（1）媒体真实性，（2）媒体背景化，（3）真实性，（4）背景覆盖。为了保留一个单一的总体可接受性标量概念，我们进一步引入一个高阶属性，称为完整性（5）。完整性表示声明是否可以被接受：低完整性表明至少有一个低阶属性被违反，而高完整性对应于对属性（2-4）的积极评估。完整性构成了主要的评估目标，而低阶属性则作为解释性因素，定位妥协的来源并简化注释任务。媒体特定属性（1-2）按媒体项目进行评估；如果没有媒体，则评估从真实性开始，针对文本部分。背景覆盖属性明确捕捉

claims that are technically true yet misleading due to omitted crucial context, a phenomenon known as “cherry-picking” (Schlichtkrull et al., 2023).

声称在技术上是正确的，但由于省略了关键背景信息而具有误导性的说法，这种现象被称为“选择性引用” (Schlichtkrull 等，2023)。

Inspired by Glockner et al. (2024), who propose to avoid a rigid class-like categorization of claims to incorporate uncertainty, we model each property on a scale from -1 to +1, where 0 denotes full uncertainty (NEI, Not Enough Information). A score of $< -1/3$ is considered a $\times$ Negative decision, analogously for $\checkmark$ Positive. The integrity score is determined directly by the worst-scoring property among (2–4), which we refer to as the compromising property. Consequently, a claim is $\checkmark$ Intact if all media (if any) are correctly contextualized, the claim veracity is true, and no false impressions arise from missing context; otherwise, it is $\times$ Compromised. If a prediction at step (2) or (3) results in a $\times$ Negative decision, the follow-up properties will not significantly impact the integrity. Thus, we terminate evaluation for the claim early for $\times$ Negative decisions at steps (2) and (3).

受 Glockner 等人（2024）的启发，他们提出避免对声明进行僵化的类比分类，以纳入不确定性，我们对每个属性进行从 -1 到 +1 的评分，其中 0 表示完全不确定性（NEI，信息不足）。得分 $< -1/3$ 被视为 $\times$ 负面决定， $\checkmark$ 正面决定类似。完整性评分由（2-4）中得分最差的属性直接决定，我们称之为妥协属性。因此，如果所有媒体（如有）都得到正确语境化，声明的真实性为真，并且没有因缺少语境而产生错误印象，则声明被视为 $\checkmark$ 完整；否则，视为 $\times$ 妥协。如果在步骤（2）或（3）中的预测结果为 $\times$ 负面决定，则后续属性将不会显著影响完整性。因此，我们将在步骤（2）和（3）中对 $\times$ 负面决定提前终止对声明的评估。

To increase robustness, we use an ensemble of four LLMs (GPT-5.2, GEMINI 2.5 PRO, CLAUDE SONNET 4.5 (Anthropic, 2025), and LLAMA 4 MAVERICK (Meta AI, 2025), aggregating predictions by their mean. We discard 318 (0.9%) claims that receive an ensemble prediction with an internal score difference exceeding 1 to keep only instances with high inter-model agreement. Additionally, each ensemble member provides a one-paragraph justification for its decision, reciting the core arguments from the fact-checking article. We instruct GPT-5 MINI to summarize the four justifications into a single one. The justification for the rating of the compromising property serves as justification for the integrity. Output. 36 K claims² with high-agreement verdicts and justifications.

为了提高鲁棒性，我们使用了四个大型语言模型（GPT-5.2、GEMINI 2.5 PRO、CLAUDE SONNET 4.5 (Anthropic, 2025) 和 LLAMA 4 MAVERICK (Meta AI, 2025)）的集成，通过取平均值来汇总预测结果。我们丢弃了 318 (0.9%) 个获得集成预测内部得分差异超过 1 的声明，只保留那些模型间高度一致的实例。此外，每个集成成员都会提供一段对其决策的说明，引用事实核查文章中的核心论点。我们指示 GPT-5 MINI 将这四个说明汇总成一个。用于评估妥协属性的评分说明也作为诚信的说明。输出。36 K 个具有高一致性裁决和说明的声明²。

3.7 Stage 7: Claim Rectification

3.7 阶段 7：索赔更正

Goal. Balance $\square$ Intact and $\square$ Compromised claims. Fact-checkers primarily verify compromised claims since these are the most harmful, cf. Fig. 17a. Only $\square$ 0.3% of ClaimReviews yield $\square$ Intact claims. Thus, we generate about 32 K “corrected” ($\square$ Intact) versions of the $\square$ Compromised claims using GPT-5.2 and GEMINI 2.5 PRO, guided by the justification of the compromising property, retaining media.

目标。平衡 $\square$ Intact 和 $\square$ Compromised 声称。事实核查员主要验证被削弱的声称，因为这是最有害的，参见图 17a。只有约 0.3% 的 ClaimReviews 会产生 $\square$ Intact 声称。因此，我们使用 GPT-5.2 和 GEMINI 2.5 PRO，根据削弱属性的合理性，保留媒体，生成大约 32,000 个“修正” ($\square$ Intact) 版本的 $\square$ Compromised 声称。

Consistency between corrected text and media is validated using GPT-5 MINI or GEMINI 2.5 FLASH. To avoid stylistic shortcuts, we use the same LLMs to validate if rectified claims are shareable, i.e., relevant, not overly specific, and worth sharing, like original claims. This way, we identify 7.3K (23.1%) rectified claims for exclusion. Finally, rectified claims are reevaluated using Stage 6. Claims with integrity $>^{1/3}$ are kept and replace their originals; all 178 (0.6%) others are discarded along with 3.1K (9.9%) claims with insufficient ensemble agreement. Output. 17 K consistent, shareable, intact rectified claims.

使用 GPT-5 MINI 或 GEMINI 2.5 FLASH 验证更正后的文本与媒体之间的一致性。为了避免风格上的简写，我们使用相同的大型语言模型（LLMs）验证更正后的声明是否可以分享，即是否相关、不过于具体且值得分享，如同原始声明。这样，我们识别出 7.3K (23.1%) 需要排除的更正后的声明。最后，使用第 6 阶段重新评估更正后的声明。具有完整性的声明 $>^{1/3}$ 会被保留并替换其原始声明；其余 178 (0.6%) 全部被丢弃，连同 3.1K (9.9%) 一致性不足的声明一并丢弃。输出。17,000 个一致、可分享且完整的更正后的声明。

4 Results

4 结果

4.1 VERITAS Statistics

4.1 VERITAS 统计数据

VERITAS currently contains 25 K claims spanning 25 quarterly splits from Q1 2020 to Q1 2026. Each quarter comprises 1 K claims, balanced so that the number of $\square$ Intact and $\square$ Compromised instances is equal. Earlier periods are excluded due to insufficient review coverage. The most recent quarter is generally the only one suitable for reliable benchmarking. In addition to the quarterly splits, we release a budget-friendly longitudinal split containing 2,500 balanced claims (100 per quarter), enabling temporal analysis across the full time span. Summary statistics are reported in Fig. 4 and App. I.4.

VERITAS 目前包含 25 个千份 (K) 索赔, 涵盖从 2020 年第一季度到 2026 年第一季度的 25 个季度划分。每个季度包含 1 个千份 (K) 索赔, 经过平衡, 使得 $\square$ 完整和 $\square$ 受损的实例数量相等。由于审查覆盖不足, 较早的时期被排除在外。最近的季度通常是唯一适合可靠基准比较的。除了季度划分之外, 我们还发布了一个经济实惠的纵向划分, 包含 2,500 个平衡的索赔 (每个季度 100 个), 从而能够对整个时间段进行时间分析。摘要统计信息在图 4 和附录 I.4 中报告。

VERITAS contains 8,692 images and 5,334 videos while it covers 54 languages, with 37 occurring at least 50 times. English is most frequent (39.0%), followed by Spanish (10.9%) and Hindi (5.8%), see Fig. 14c for more languages. Claim appearances are predominantly sourced from Facebook (38.4%) and X/Twitter (20.5%), while all other platforms contribute less than 6% each. AFP Fact Checking accounts for the largest share of reviewed claims (24.8%), with all other organizations contributing below 6% each.

VERITAS 包含 8,692 张图片和 5,334 个视频, 覆盖 54 种语言, 其中 37 种语言至少出现 50 次。英语出现频率最高 (39.0%), 其次是西班牙语 (10.9%) 和印地语 (5.8%), 更多信息请参见图 14c。声明的来源主要来自 Facebook (38.4%) 和 X/Twitter (20.5%), 而其他所有平台各自的贡献均低于 6%。AFP 事实核查账户在审查的声明中占比最大 (24.8%), 其他所有组织的贡献均低于 6%。

4.2 Validation through Human Evaluation

4.2 通过人工评估进行验证

We validate the outputs from stages 6 and 7 (Sec. 3.6) via human evaluation. Each claim is independently annotated by ≥ 2 native or C1-level speakers of the claim's language. Annotators sequentially assess all properties, following the exact same annotation procedure as in stage 6, including scoring and providing written justifications.

我们通过人工评估验证第 6 和第 7 阶段 (第 3.6 节) 的输出。每个声明由 ≥ 2 母语为该声明语言的原住民或 C1 水平的说话者独立标注。标注人员依次评估所有属性, 遵循与第 6 阶段完全相同的标注流程, 包括评分和提供书面理由。

We report Mean Squared Error (MSE), Mean Absolute Error (MAE), and accuracy after discretizing scores into three or seven bins, respectively. MSE is our primary metric, as it penalizes severe confusions more strongly (App. D). MSE $\square$ 0.04 indicates very high agreement (App. D). Results are

我们报告均方误差 (MSE)、平均绝对误差 (MAE) 以及将评分离散化为三个或七个区间的准确率。MSE 是我们的主要指标, 因为它对严重的混淆惩罚更强烈 (附录 D)。MSE $\square$ 0.04 表示非常高的一致性 (附录 D)。结果为

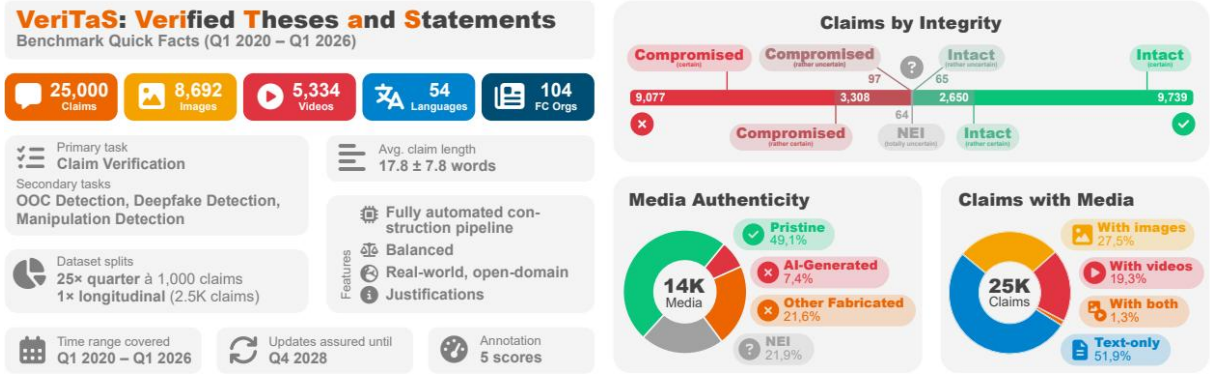


Figure 4: VERITAS dataset statistics, see also Tab. 8.

图 4: VERITAS 数据集统计信息，参见表 8。

Integrity	N	Error Rates ($\square$)		Accuracy ($\square$)	
MSE	MAE	7-bin	3-bin		
VERITAS	204	0.034	0.102	69.1	97.5
- w/ ensemble, w/o filtering	207	0.035	0.105	68.6	97.6
- GPT-5.2 alone	207	0.076	0.184	51.2	95.7
- GEMINI 3.1 PRO alone	207	0.071	0.099	73.4	95.7
- CLAUDE SONNET 4 alone	207	0.048	0.091	72.5	97.1
- LLAMA 4 MAVERICK alone	207	0.042	0.103	66.7	97.1

诚信	N	错误率 ($\square$)		准确度 ($\square$)	
MSE	MAE	7-bin	3-bin		
VERITAS	204	0.034	0.102	69.1	97.5
- 带集成，不带过滤	207	0.035	0.105	68.6	97.6
- GPT-5.2 单独使用	207	0.076	0.184	51.2	95.7
- GEMINI 3.1 PRO 单独使用	207	0.071	0.099	73.4	95.7
- CLAUDE SONNET 4 alone	207	0.048	0.091	72.5	97.1
- LLAMA 4 MAVERICK 单独使用	207	0.042	0.103	66.7	97.1

Table 2: Human evaluation results over N claims. The scores reflect agreement between human annotation and automated VERITAS annotations for integrity for six different settings. MSE = Mean Squared Error (the main metric), MAE = Mean Absolute Error, Accuracy = share of matches of n-bin discretized scores (in %).

表 2: 人类评估结果在 N 个声明上的表现。分数反映了人类标注与自动化 VERITAS 标注在六个不同设置下关于完整性的共识。MSE = 均方误差（主要指标），MAE = 平均绝对误差，Accuracy = n-bin 离散分数匹配的比例（以百分比表示）。

summarized in Tab. 2, where we also include five ablated VERITAS variants: w/ ensemble, w/o filtering (keeping claims with high ensemble disagreement) and just one of the four ensemble members as the sole predictor. The evaluation shows a very strong agreement with an MSE below 0.04, corresponding to roughly one flipped prediction among 100 otherwise correct judgments. The observations are confirmed by confusion matrices (Fig. 25) and the results for low-level properties (Tab. 10).

总结在表 2 中，其中我们还包含了五个被削弱的 VERITAS 变体：使用集成、不进行过滤（保留高集成分歧的声明）以及仅使用四个集成成员中的一个作为唯一预测器。评估结果显示了非常强的一致性，均方误差（MSE）低于 0.04，对应于在 100 个其他正确的判断中大约有一个预测被翻转。这些观察结果得到了混淆矩阵（图 25）以及低级属性结果（表 10）的证实。

Overall, these results confirm that the automated verdict mapping closely aligns with human judgments, validating the VERITAS construction pipeline. Additionally, the increased error rates for the ablated variants in Tab. 2 indicate the benefits of the ensemble approach and the agreement filtering. Moreover, evaluators were asked to flag problematic claims. Only a small minority ($\sim 5.1\%$) exhibited quality concerns (primarily missing media, otherwise clarity issues), indicating that the normalized/rectified claims are generally well-formed and realistic. See App. K for more details on the evaluation.

总体而言，这些结果证实了自动判决映射与人类判断高度一致，验证了 VERITAS 构建流程的有效性。此外，表 2 中去除某些组件后的变体错误率增加，表明了集成方法和一致性过滤的优势。此外，评估人员被要求标记有问题的声明。仅有少数 ($\sim 5.1\%$) 表现出质量方面的担忧（主要缺失媒体，否则是清晰度问题），这表明标准化/修正后的声明通常结构良好且现实。有关评估的更多细节，请参见附录 K。

4.3 Benchmarking AFC Methods on VERITAS latest quarter

4.3 在 VERITAS 最新季度对 AFC 方法进行基准测试

We analyze baselines and current AFC methods on the Q1 2026 split to assess performance on the most recent data—particularly after the knowledge cutoff of all tested methods. We consider various foundation models: CLAUDE OPUS 4.6, GPT-5.2 and GEMINI 3.1 PRO as state-of-the-art multimodal models; GPT-4O as an earlier-generation model to study knowledge-cutoff effects; GEMINI 3 FLASH as a mid-sized LLM, as well as LLAMA 4 MAVERICK, GEMMA 4 (31B), and QWEN 3.5 (397B) as representative open-source models. Each model is evaluated once using parametric knowledge only and once with web search augmentation, see App. J.1 for the tool implementation details. Note that all evaluations on VERITAS require excluding evidence sources published after the claim’s release date to avoid temporal leakage (Glockner et al., 2022).

我们分析基线和当前 AFC 方法在 2026 年第一季度分割数据集上的表现，以评估在最近数据上的性能——特别是所有测试方法的知识截止点之后。我们考虑了各种基础模型：CLAUDE OPUS 4.6、GPT-5.2 和 GEMINI 3.1 PRO 作为最先进的多模态模型；GPT-4O 作为早期一代模型以研究知识截止效应；GEMINI 3 FLASH 作为中型 LLM，以及 LLAMA 4 MAVERICK、GEMMA 4 (31B) 和 QWEN 3.5 (397B) 作为代表性开源模型。每个模型均使用仅参数知识进行一次评估，并使用网络搜索增强进行另一次评估，详见附录 J.1 中的工具实现细节。请注意，所有在 VERITAS 上的评估都需要排除在声明发布日期之后发布的证据来源，以避免时间泄露（Glockner 等，2022）。

GEMINI 3.1 PRO and GEMINI 3 FLASH process videos natively. For all other models, we represent videos via five evenly spaced frames and a speech transcript. Additionally, we compare against two recent fact-checking systems, DEFAME (Braun et al., 2025) and LOKI (Li et al., 2025b), both with three different backbones in Tab. 3.

GEMINI 3.1 PRO 和 GEMINI 3 FLASH 可以原生处理视频。对于所有其他模型，我们通过五个等间距的帧和语音转录文本来表示视频。此外，我们还与两个最近的事实核查系统进行比较，DEFAME (Braun 等，2025) 和 LOKI (Li 等，2025b)，两者在表 3 中均使用了三种不同的骨干网络。

Across all baselines, CLAUDE OPUS 4.6 achieves the strongest MSE in both evaluation settings, even surpassing the AFC-specialized models DEFAME and LOKI. The second-best model, GEMINI 3 FLASH, even surpassing GEMINI 3.1 PRO, trails by +0.18 and +0.09 MSE, depending on retrieval access. Open-source LLAMA 4 MAVERICK performs the worst among LLMs, also under search augmentation. In contrast, open-source GEMMA 4 (31B) and QWEN 3.5 (397B) both perform on par or better

在所有基线模型中，CLAUDE OPUS 4.6 在两种评估设置中均实现了最强的 MSE，甚至超过了专门针对

AFC 的模型 DEFAME 和 LOKI。第二好的模型 GEMINI 3 FLASH，即使超过了 GEMINI 3.1 PRO，在检索访问方面也分别落后 +0.18 和 +0.09 MSE。开源 LLAMA 4 MAVERICK 在 LLMs 中表现最差，即使在搜索增强的情况下也是如此。相比之下，开源 GEMMA 4 (31B) 和 QWEN 3.5 (397B) 的表现均与其它模型相当或更优。

Method		Search Overall Results					MSE on Subsets (□)					
Errors (□)		Accuracy (□)		By Media			By Language	By Integrity				
MSE	MAE	7-bin	3-bin	w/ images	w/ videos	text-only	English	non-English	Intact	Compr.		
GEMINI 3 FLASH		0.632	0.409	60.3	81.4	0.553	0.743	0.625	0.604	0.648	1.097	0.167
GEMINI 3.1 PRO		0.636	0.412	58.9	81.1	0.508	0.751	0.659	0.617	0.647	1.210	0.049
CLAUDE OPUS 4.6		0.453	0.471	30.1	62.2	0.451	0.509	0.361	0.450	0.455	0.572	0.337
GPT-4O	-	0.826	0.690	25.4	45.3	1.050	0.759	0.560	0.967	0.760	1.418	0.247
GPT-5.2	-	0.701	0.691	12.8	42.0	0.644	0.820	0.571	0.678	0.710	1.179	0.240
LLAMA 4 MAVERICK		0.943	0.767	18.1	44.2	1.127	0.949	0.583	0.966	0.931	1.270	0.635
GEMMA 4 (31B)		0.675	0.465	51.5	77.2	0.590	0.791	0.703	0.632	0.697	1.059	0.290
QWEN 3.5 (397B)		0.887	0.563	51.3	69.8	0.828	1.070	0.587	0.911	0.881	1.404	0.373
GEMINI 3 FLASH		0.275	0.237	67.2	90.5	0.250	0.332	0.158	0.268	0.279	0.371	0.173
GEMINI 3.1 PRO		0.388	0.316	61.1	82.9	0.335	0.443	0.277	0.397	0.386	0.643	0.132
CLAUDE OPUS 4.6		0.183	0.221	60.8	89.6	0.134	0.210	0.196	0.172	0.188	0.161	0.202
GPT-4O	□	0.841	0.615	38.4	59.6	1.141	0.702	0.547	0.993	0.770	1.362	0.333
GPT-5.2	□	0.353	0.422	22.8	77.1	0.271	0.437	0.313	0.261	0.393	0.544	0.166

LLAMA 4	0.863	0.712	22.0	50.2	0.934	0.904	0.535	0.811	0.878	1.032	0.707
MAV-ER-ICK											
GEMMA 4 (31B)	0.360	0.295	60.7	87.2	0.328	0.429	0.236	0.331	0.374	0.277	0.438
QWEN 3.5 (397B)	0.318	0.296	57.8	86.7	0.263	0.354	0.254	0.237	0.354	0.393	0.240
DEFAME w/ GEMINI 3 FLASH	0.450	0.320	62.2	87.2	0.318	0.589	0.319	0.387	0.472	0.507	0.391
DEFAME w/ CLAUDE OPUS 4.6	0.282	0.277	58.1	88.3	0.157	0.411	0.283	0.213	0.314	0.440	0.123
DEFAME w/ GEMMA 4 (31B)	0.465	0.349	59.3	83.6	0.344	0.599	0.316	0.329	0.527	0.491	0.441
LOKI w/ GEMINI 3 FLASH	1.487	0.852	39.4	54.8	1.422	1.541	1.659	1.295	1.561	2.821	0.151
LOKI w/ CLAUDE OPUS 4.6	1.638	0.920	38.0	52.0	1.648	1.797	1.359	1.657	1.623	3.195	0.080
LOKI w/ GEMMA 4 (31B)	1.109	0.800	23.0	42.5	1.025	1.214	0.894	0.871	1.215	0.892	1.336

方法	搜索	总体结果	子集上的均方误差 (□)
错误 (□)		准确度 (□)	按媒体 按语言 以诚信为本

MSE	MAE	7- bin	三分 类	带图 片	带视 频	纯文 本	中文	非英 语	完整	压缩		
GEMINI 3 FLASH		0.632	0.409	60.3	81.4	0.553	0.743	0.625	0.604	0.648	1.097	0.167
GEMINI 3.1 PRO		0.636	0.412	58.9	81.1	0.508	0.751	0.659	0.617	0.647	1.210	0.049
CLAUDE OPUS 4.6		0.453	0.471	30.1	62.2	0.451	0.509	0.361	0.450	0.455	0.572	0.337
GPT- - 4O		0.826	0.690	25.4	45.3	1.050	0.759	0.560	0.967	0.760	1.418	0.247
GPT- - 5.2		0.701	0.691	12.8	42.0	0.644	0.820	0.571	0.678	0.710	1.179	0.240
LLAMA 4 MAV- ER- ICK		0.943	0.767	18.1	44.2	1.127	0.949	0.583	0.966	0.931	1.270	0.635
GEMMA 4 (31B)		0.675	0.465	51.5	77.2	0.590	0.791	0.703	0.632	0.697	1.059	0.290
QWEN - 3.5 (397B)		0.887	0.563	51.3	69.8	0.828	1.070	0.587	0.911	0.881	1.404	0.373
GEMINI 3 FLASH		0.275	0.237	67.2	90.5	0.250	0.332	0.158	0.268	0.279	0.371	0.173
GEMINI 3.1 PRO		0.388	0.316	61.1	82.9	0.335	0.443	0.277	0.397	0.386	0.643	0.132
CLAUDE OPUS 4.6		0.183	0.221	60.8	89.6	0.134	0.210	0.196	0.172	0.188	0.161	0.202
GPT- □ 4O		0.841	0.615	38.4	59.6	1.141	0.702	0.547	0.993	0.770	1.362	0.333
GPT- □ 5.2		0.353	0.422	22.8	77.1	0.271	0.437	0.313	0.261	0.393	0.544	0.166
LLAMA□ 4 MAV- ER- ICK		0.863	0.712	22.0	50.2	0.934	0.904	0.535	0.811	0.878	1.032	0.707
GEMMA□ 4 (31B)		0.360	0.295	60.7	87.2	0.328	0.429	0.236	0.331	0.374	0.277	0.438
QWEN □ 3.5 (397B)		0.318	0.296	57.8	86.7	0.263	0.354	0.254	0.237	0.354	0.393	0.240

DEFAME w/ GEM- INI 3 FLASH		0.450	0.320	62.2	87.2	0.318	0.589	0.319	0.387	0.472	0.507	0.391
DEFAME w/ CLAUDE OPUS 4.6		0.282	0.277	58.1	88.3	0.157	0.411	0.283	0.213	0.314	0.440	0.123
DEFAME w/ GEMMA 4 (31B)		0.465	0.349	59.3	83.6	0.344	0.599	0.316	0.329	0.527	0.491	0.441
洛基 □ w/ 双子 座 3 闪电		1.487	0.852	39.4	54.8	1.422	1.541	1.659	1.295	1.561	2.821	0.151
洛基 □ w/ 克劳 德 OPUS 4.6		1.638	0.920	38.0	52.0	1.648	1.797	1.359	1.657	1.623	3.195	0.080
洛基 □ w/ 蓬塔 4 (31B)		1.109	0.800	23.0	42.5	1.025	1.214	0.894	0.871	1.215	0.892	1.336

Table 3: Model evaluation on the Q1 2026 split with(out) search tool, single runs, with best and second best.

表 3：在 2026 年第一季度划分上模型评估，使用（不使用）搜索工具，单次运行，包含最佳和次佳结果。

than top-tier GEMINI 3.1 PRO. Surprisingly, despite its specialization, LOKI underperforms all other models. Subset results reveal several observations: (1) Video claims are the most difficult to fact-check for almost all models, (2) only GPT-5.2 and QWEN 3.5 (397B) seem to noticeably struggle with non-English claims, and (3) many models exhibit a strong bias towards rating claims as □ Compromised.

比顶级的 GEMINI 3.1 PRO 更强大。令人惊讶的是，尽管 LOKI 有专门的训练，它在所有其他模型中的表现却都不理想。子集结果揭示了几个观察点：(1) 视频声明对于几乎所有模型来说都是最难进行事实核查的，(2) 似乎只有 GPT-5.2 和 QWEN 3.5 (397B) 在处理非英语声明时明显存在困难，(3) 许多模型表现出对将声明标记为 □ 有偏见。

4.4 The Role of Knowledge Cutoff

4.4 知识截止的作用

We evaluate the same models on the longitudinal VERITAS split. Fig. 5 shows the results. All evaluated models exhibit a marked increase in MSE after their respective Knowledge Cutoff Dates (KCDs), as one can see for the consistently increasing solid curves. On average, MSE rises from roughly 0.6 to above 0.8, substantially reducing the margin to trivial always-NEI behavior. The effect is particularly pronounced for GEMINI 3.1 PRO, whose MSE increases from around 0.3 to over 0.7. We can rule out the increasing share of videos as a confounding factor, since the same phenomenon is observed for text-only claims, see Fig. 22 for the breakdown.

我们在纵向的 VERITAS 分割上评估了相同的模型。图 5 展示了结果。所有评估的模型在其各自的知识截止日期（KCDs）之后都表现出明显的均方误差（MSE）上升，这一点可以从不断上升的实线曲线中看出。平均而言，MSE 从大约 0.6 上升到超过 0.8，大大缩小了与平凡的始终-NEI 行为之间的差距。这种效应在 GEMINI 3.1 PRO 中尤为明显，其 MSE 从大约 0.3 上升到超过 0.7。我们可以排除视频数量增加作为混杂因素，因为对于纯文本声明也观察到了相同的现象，参见图 22 的分解。

4.5 Real-World Insights

4.5 实际应用见解

The data obtained from stages 1–6 roughly reflect the natural distribution of real-world fact-checked claims and reveal significant temporal trends, despite the dismissal of numerous reviews and claims. The share of AI-generated media steadily increases

从阶段 1 到 6 获得的数据大致反映了现实世界中经过核实的声明的自然分布，并揭示了显著的时间趋势，尽管许多评论和声明被驳回。人工智能生成的媒体所占的比例稳步上升

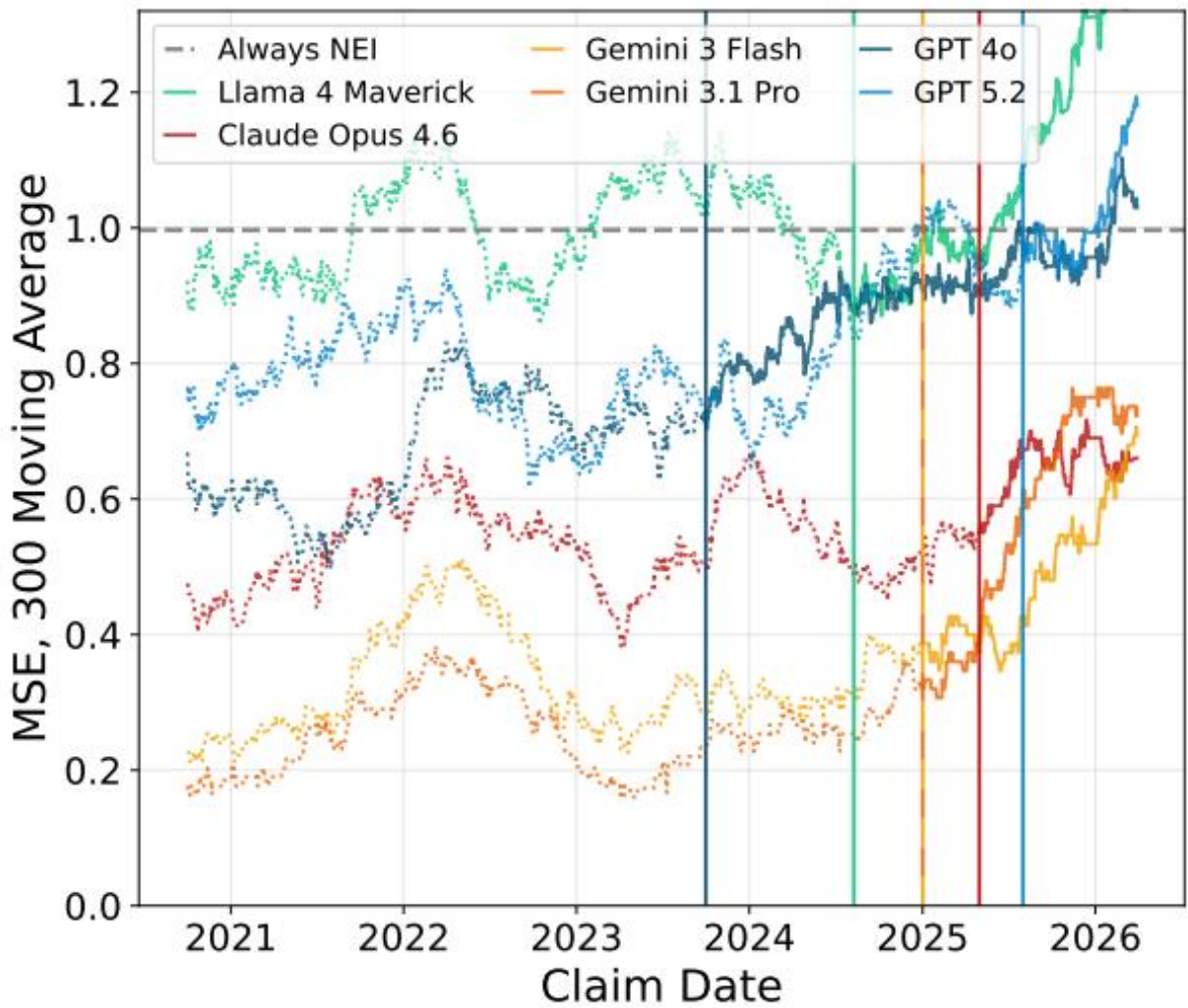


Figure 9: images/1969010cc29abc061c44dd3bda08219035a322b9f82a320332e3e439c3ba7d70.jpg

Figure 5: Baseline MSE performance without web search on the longitudinal split, single runs, smoothed with moving average by claim date. Lower is better. Vertical lines indicate knowledge cutoff dates, solid curves post-cutoff performance.

图 5：在纵向分割上，不进行网络搜索的基线 MSE 性能，单次运行，按索赔日期进行移动平均平滑处理。数值越低越好。垂直线表示知识截止日期，实线表示截止日期后的性能表现。

over time, exceeding 25% of claims by Q4 2025, while the proportion of pristine media correspondingly declines (Fig. 18). The fraction of claims containing media—particularly videos—also increases over time (Fig. 17). Platform coverage shifts are observable, with a declining share of Facebook and a growing share of Instagram (Fig. 18). Finally, the volume of all published ClaimReviews rises until Q1 2025 and decreases thereafter (Fig. 9).

随着时间的推移，超过 25% 的声明在 2025 年第四季度之前出现，而原始媒体的比例相应下降（图 18）。包含媒体（尤其是视频）的声明比例也随着时间的推移而增加（图 17）。平台覆盖范围发生了变化，Facebook 的占比下降，而 Instagram 的占比上升（图 18）。最后，所有已发布 ClaimReviews 的数量直到 2025 年第一季度持续上升，之后则开始下降（图 9）。

5 Discussion

5 讨论

Data leakage severely undermines static AFC benchmarks. The increase in error rate of models after their Knowledge Cutoff Date (KCD) provides strong empirical evidence that LLMs partially encode real-world claims and verdicts from before their KCD. Vice versa, benchmarks dominated by pre-KCD data systematically overestimate model performance. This “pre-cutoff inflation” phenomenon is concerning since realistic fact-checking usually targets newly emerging claims. Notably, the most recent KCD among evaluated models (GPT-5.2: Aug 31, 2025) postdates a large fraction of existing AFC benchmarks, rendering their evaluation increasingly unreliable.

数据泄露严重削弱了静态 AFC 基准。模型在其知识截止日期（KCD）之后错误率的增加，提供了强有力的实证证据，表明 LLMs 在某种程度上编码了其 KCD 之前现实世界中的主张和判决。相反，以预 KCD 数据为主的基准系统性地高估了模型的性能。这种“预截止膨胀”现象令人担忧，因为现实中的事实核查通常针对新出现的主张。值得注意的是，评估模型中最近期的 KCD（GPT-5.2：2025 年 8 月 31 日）晚于现有 AFC 基准的很大一部分，使得其评估结果越来越不可靠。

Nature and Validity of Rectified Claims. High-quality AFC benchmarks should be grounded in real claims, as purely synthetic data fails to capture the diversity and realism of misinformation and raises ethical concerns. In contrast, VERITAS generates $\square$ Intact claims conditioned on expert ground truth through rectification. Although roughly 50% of claims are generated this way, they are fundamentally different from synthetic claims in prior work. VERITAS’ rectified claims are directly grounded in real-world fact-checking articles and their associated evidence, rather than being artificially constructed or designed to mislead. Moreover, they undergo a multi-stage validation process, including filtering, where a substantial fraction is discarded, as well as an additional Integrity check. Finally, human evaluation indicates that only a small minority of all claims ($\sim 5.1\%$) exhibit quality concerns. Taken together, these results suggest that rectified claims are not synthetic artifacts in the conventional sense, but rather evidence-grounded corrections of real-world claims that preserve realism while enabling dataset balancing.

修正声明的性质与有效性。高质量的 AFC 基准应基于真实的声明，因为纯粹的合成数据无法捕捉虚假信息多样性和现实性，并且会引发伦理问题。相反，VERITAS 通过修正生成 $\square$ 完整声明，这些声明基于专家的地面真相。尽管大约 50% 的声明是通过这种方式生成的，但它们与之前工作中合成声明的根本不同。VERITAS 的修正声明直接基于现实世界的事实核查文章及其相关证据，而不是人工构造或设计误导性的声明。此外，它们还经过多阶段的验证过程，包括过滤阶段，其中大量声明被舍弃，以及额外的完整性检查。最后，人工评估表明，所有声明中只有极少数（ $\sim 5.1\%$ ）存在质量问题。综上所述，这些结果表明，修正声明在传统意义上并不是合成的产物，而是基于证据对现实世界声明的更正，这些更正在保持现实性的同时，使数据集得以平衡。

Substantial room for improvement. No evaluated model approaches an MSE of 0.1, which we consider a reasonable threshold for acceptable AFC. Even the best-performing model (CLAUDE OPUS 4.6, MSE = 0.18) corresponds to approximately one flipped prediction in 22 otherwise perfect ratings. Specialized AFC systems appear to degrade the performance of the tested LLMs when used as backbones. These results underscore that multimodal AFC remains far from solved. Finally, the fact that GEMMA 4 (31B) performs on par with substantially larger proprietary models such as GPT-5.2 and GEMINI 3.1 PRO suggests that model scale alone is not required to achieve decent performance.

有很大的改进空间。目前没有评估的模型能够达到 0.1 的 MSE，我们认为这个阈值对于可接受的 AFC 是合理的。即便是表现最好的模型（CLAUDE OPUS 4.6, MSE = 0.18）也对应于 22 个其他完美评分中的大约一个预测错误。专用的 AFC 系统在作为骨干网络使用时似乎会降低测试的 LLMs 的性能。这些结果表明，多模态 AFC 仍然远未解决。最后，GEMMA 4（31B）的表现与诸如 GPT-5.2 和 GEMINI 3.1 PRO 等明显更大的专有模型相当，这表明仅靠模型规模并不需要实现良好的性能。

Reliance on Proprietary LLMs. Dependence on proprietary models raises concerns regarding reproducibility. To assess sensitivity to the underlying backbone, we partially regenerated a subset of Q4 2025 using a substantially different LLM family (see App. H). The resulting claims

preserve the same core assertions with only minor wording differences, and ensemble-based verdicts remain highly consistent. While this does not yet fully demonstrate reproducibility with open-weight models, these findings suggest that VERITAS is largely invariant to the specific proprietary backbone.

依赖专有 LLMs。对专有模型的依赖引发了关于可重复性的担忧。为了评估对底层骨干模型的敏感性，我们使用与之前显著不同的 LLM 家族部分重新生成了 2025 年第四季度的一小部分数据（参见附录 H）。由此产生的声明保持了相同的中心论点，仅在措辞上略有不同，基于集成的判决仍然高度一致。虽然这尚未完全证明与开放权重模型的可重复性，但这些发现表明 VERITAS 在很大程度上与特定的专有骨干模型无关。

Outlook. The dynamic design of VERITAS will capture emerging misinformation trends through quarterly updates. Future work must address potential paradigm shifts, such as increased prevalence of audio-only content or platform restrictions that limit content access. Scaling claim retrieval beyond the current 25 K claims toward the full set of roughly 400 K reviews would further improve coverage and analytical power. Since claims can be ambiguous, future work should find strategies to address multiple interpretations of claims. Another avenue of research is required to incorporate the claimant’s intent: Satirical content, while most often being “Compromised”, could be considered “Intact”, given its humorous intention. Finally, since VERITAS only evaluates verdict scores, future work would need to develop evaluation approaches for the presented evidence.

展望。VERITAS 动态的设计将通过季度更新来捕捉新兴的错误信息趋势。未来的工作必须应对潜在的范式转变，例如音频内容的普遍增加或限制内容访问的平台限制。将声明检索扩展到当前 25,000 条声明之外，覆盖大约 400,000 条评论的完整集合，将进一步提高覆盖率和分析能力。由于声明可能具有歧义，未来的工作应寻找应对声明多种解释的策略。另一个研究方向是纳入声明者的意图：讽刺性内容虽然通常被标记为“受损”，但考虑到其幽默意图，也可以被视为“完整”。最后，由于 VERITAS 仅评估判决分数，未来的工作需要开发用于呈现证据的评估方法。

6 Conclusion

6 结论

We introduced VERITAS, the first dynamic benchmark for multimodal automated fact-checking, designed to remain robust under continual large-scale pretraining of foundation models. By collecting real-world, multilingual, multimodal claims via a fully automated pipeline and updating the benchmark quarterly, VERITAS mitigates data leakage that increasingly invalidates static AFC benchmarks. Human evaluation validates both the annotation scheme and the construction process. Baseline experiments reveal that current state-of-the-art multimodal LLMs fall far short of reliable fact-checking performance, while performance on pre-knowledge-cutoff data substantially overestimates real-world capability. VERITAS establishes a principled foundation for meaningful, future-proof evaluation of multimodal AFC systems.

我们介绍了 VERITAS，这是首个用于多模态自动事实核查的动态基准，旨在在基础模型持续大规模预训练下保持稳健性。通过完全自动化的流程收集真实世界、多语言、多模态的声明，并每季度更新基准，VERITAS 减少了数据泄露问题，而数据泄露正越来越多地使静态 AFC 基准失效。人工评估验证了标注方案和构建过程。基线实验表明，当前最先进的多模态大语言模型在可靠的事实核查性能上远远不足，而基于预知识截止点之前数据的性能则显著高估了实际能力。VERITAS 为多模态 AFC 系统的有意义、未来可靠的评估奠定了原则性基础。

7 Limitations

7 局限性

Despite its scope and level of automation, VERITAS has several limitations. First, we do not perform rigorous cross-lingual claim deduplication. While the share of duplicates in the release data appears to be very low (App. G), semantically equivalent claims appear across languages, particularly when fact-checking organizations publish parallel versions of the same claim (e.g., AFP). While this reflects real-world dissemination, it introduces a minor redundancy within splits. Future VERITAS iterations should introduce textual and visual claim deduplication for sampling.

尽管 VERITAS 在范围和自动化程度方面都有所涵盖，但它仍然存在一些限制。首先，我们并未进行严格的跨语言声明去重。虽然在发布数据中重复项的比例似乎非常低（附录 G），但在不同语言中会出现语义等价的声明，尤其是在事实核查机构发布同一声明的平行版本时（例如 AFP）。虽然这反映了现实中的传播情况，但会在数据集中引入轻微的冗余。未来 VERITAS 的版本应引入文本和视觉声明去重机制以进行抽样。

Second, although rectified claims are generated using targeted prompting and automated validation, their linguistic style is not always optimal. Some rectified claims may remain verbose or insufficiently natural, which could expose exploitable shortcuts for inference. While analysis shows no noticeable advantage on rectified claims (App. F), explicit prompting or finetuning strategies might still exploit shortcuts.

第二，尽管使用定向提示和自动验证生成了修正后的声明，但它们的语言风格并不总是最佳的。一些修正后的声明可能仍然冗长或不够自然，这可能会暴露出可用于推理的捷径。虽然分析显示在修正后的声明上没有明显的优点（附录 F），但显式提示或微调策略仍可能利用这些捷径。

Third, VERITAS is designed around current modalities, platforms, and publication practices. Potential paradigmatic shifts—such as increased prevalence of audio-only content, emergence of new communication platforms, restrictions on media download, or the discontinuation of fact-checking infrastructures, e.g., ClaimReview by Google or Data Commons or fact-checking organizations themselves—could impair future data collection. While the pipeline has operated reliably for data from the past six years and successfully demonstrated its extension in Q1 2026, its long-term robustness cannot be guaranteed.

第三，VERITAS 是围绕当前的模式、平台和出版实践设计的。潜在的范式转变——例如纯音频内容的普及率增加、新通信平台的出现、媒体下载的限制，或事实核查基础设施的终止（例如 Google 的 ClaimReview、Data Commons 或事实核查组织本身）——可能会妨碍未来数据的收集。虽然该数据管道在过去六年中一直可靠运行，并在 2026 年第一季度成功展示了其扩展能力，但其长期的稳健性无法保证。

Finally, although fully automated, the construction of VERITAS incurs non-trivial API and computational costs. Creating the full benchmark required approximately \$16.5K so far, corresponding to about \$660 per quarter split. Additionally, hosting LLAMA 4 MAVERICK in-house required roughly 2.9K GPU hours on 8 NVIDIA H100 GPUs, i.e., about 116 h per quarter. Note that these costs do not affect users of VERITAS, which is freely available at veritas.mai.informatik.tu-darmstadt.de.

最后，尽管 VERITAS 是全自动化的，但其构建仍涉及非微不足道的 API 和计算成本。创建完整的基准测试需要大约每个季度约 \$16.5K so far, corresponding to about \$660。此外，内部托管 LLAMA 4 MAVERICK 大约需要 8 块 NVIDIA H100 GPU 上的 2900 小时 GPU 计算时间，即每个季度约 116 小时。请注意，这些成本不会影响 VERITAS 的用户，VERITAS 可在 veritas.mai.informatik.tu-darmstadt.de 免费获取。

8 Ethical Considerations

8 道德考量

With the introduction of VERITAS, we aim to support society in addressing mis- and disinformation by enabling more reliable and meaningful evaluation of multimodal AFC systems. By solving issues of previous benchmarks, VERITAS is intended to accelerate responsible AFC research sustainably.

随着 VERITAS 的引入，我们的目标是通过使多模态 AFC 系统的评估更加可靠和有意义，支持社会应对错误信息和虚假信息。通过解决先前基准测试中的问题，VERITAS 旨在可持续地加速负责任的 AFC 研究。

VERITAS relies exclusively on real-world claims that have already been fact-checked by professional organizations. This design choice mitigates ethical risks associated with generating synthetic misinformation, which could otherwise contribute to the creation or dissemination of harmful content. Using expert-derived judgments promotes transparency, accountability, and reproducibility, while preserving the realism necessary for valid evaluation.

VERITAS 仅依赖于已经由专业机构核实的现实世界中的声明。这种设计选择可以减轻生成合成虚假信息所带来的伦理风险，否则这些信息可能会助长有害内容的创建或传播。使用专家得出的判断有助于提高透明度、问责制和可重复性，同时保持有效评估所需的现实感。

At the same time, potential misuse risks exist. Models evaluated on VERITAS, when used to monitor social networks, could be repurposed for surveillance, censorship, or selective moderation, particularly in politically sensitive contexts. Moreover, although VERITAS reduces evaluation bias caused by data leakage, it may still reflect geographic, linguistic, or institutional biases inherent in the global fact-checking ecosystem.

同时，潜在的误用风险依然存在。在 VERITAS 上评估的模型，当用于监控社交网络时，可能会被重新用于监视、审查或选择性管理，尤其是在政治敏感的环境中。此外，尽管 VERITAS 通过减少数据泄露导致的评估偏差，但它仍可能反映出全球事实核查生态系统中固有的地理、语言或机构偏见。

All data are sourced from publicly available materials; therefore, we did not anonymize individual claims or media. However, since some content may have been removed from the web after our collection, for copyright compliance, and to avoid automated scraping, access to VERITAS is restricted to eligible researchers via a gated request process. Finally, the dataset may contain offensive or disturbing content, which we intentionally retain, as such material is an intrinsic part of real-world fact-checking and necessary for faithful evaluation.

所有数据均来源于公开资料，因此我们未对个人声明或媒体内容进行匿名化处理。然而，由于在我们收集数据之后，某些内容可能已被从网络上删除，为遵守版权规定并避免自动化抓取，只有通过受限请求流程的合格研究人员才能访问 VERITAS。最后，该数据集可能包含冒犯性或令人不安的内容，我们有意保留这些内容，因为此类材料是现实世界事实核查的一部分，也是进行忠实评估所必需的。

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